Chapter 3. Permutation Tests

Randomization and population models, conditional inference, permutation tests for common problems

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3.1 Two sources of randomness

- **Randomization model.** N fixed units, m randomly assigned to treatment 1 and $n = N - m$ to treatment 2; responses are fixed, only the assignment is random.
- **Population model.** $X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are random samples. Inference concerns F and G, e.g. $H_0 : F = G$.

Both use the *permutation* (*randomization*) distribution of a statistic over all re-assignments of the observed values.

3.2 The randomization test

Under H_0 (no treatment effect) each response is the same under either treatment, and the observed assignment is one of $K = \binom{N}{m}$ equally likely ones, all yielding the same pooled responses. With $d_1, \dots, d_K$ the values of a statistic D (e.g. $\bar{Y} - \bar{X}$) over all assignments,

$$p = \frac{1}{K} \sum_{k=1}^{K} I(d_k \geq D_{\text{obs}}). \tag{3.1}$$

Two-sided: use $|d_k| \geq |D_{\text{obs}}|$, not twice the smaller one-sided value (the distribution need not be symmetric).

Example 3.1

Five patients; two random ones get a new treatment (recovery times $3, 5$), the others have $8, 9, 12$. With $D = \bar{Y}_{\text{old}} - \bar{X}_{\text{new}}$ and pooled total 37, a treated pair with sum s gives $D = (37 - s)/3 - s/2$, decreasing in s. The 10 pair sums are $8, 11, 12, 13, 14, 15, 17, 17, 20, 21$; the observed 8 is smallest, so $D_{\text{obs}} = 5.67$ is largest and $p = 1/10$.

Proposition 3.1 (Exactness)

Under H_0, if the d_k have no ties and $\alpha = j/K$, the randomization test rejecting when $p \leq \alpha$ has type I error exactly α. With ties it is at most α.

Proof. Under H_0 the multiset $\{d_k\}$ does not depend on the realized assignment, which is uniform, so D_{obs} is a uniform element of it. Without ties, $p \leq j/K$ iff D_{obs} is among the j largest, probability j/K; ties only increase p. ■

Proposition 3.2 (Equivalent statistics)

Over all re-assignments of a fixed pooled sample, $\sum_j Y_j$, $\bar{Y} - \bar{X}$ and the pooled two-sample t-statistic are strictly increasing functions of each other, hence give identical permutation p-values.

Proof. The pooled total $\sum X_i + \sum Y_j$ is fixed, so $\bar{Y} - \bar{X} = \frac{N}{mn} \sum Y_j - \frac{\text{total}}{m}$ is increasing in $\sum Y_j$. The total sum of squares SST is also fixed, and $SST = (N-2)s_p^2 + \frac{mn}{N} D^2$ with $D = \bar{Y} - \bar{X}$. Hence

$$T = \frac{D}{\sqrt{\frac{1}{m} + \frac{1}{n}} \sqrt{(SST - \frac{mn}{N} D^2)/(N-2)}} = g(D), \qquad g'(D) \propto \frac{SST}{(SST - \frac{mn}{N} D^2)^{3/2}} > 0.$$

■

A difference of medians or trimmed means is *not* monotone in D and can give different (for skewed or heavy-tailed data often more powerful) tests.

3.3 Permutation tests in the population model

Motivation. Under $H_0 : F = G$ the pooled sample is exchangeable, so permutation tests remain valid for random samples.

Definition (Exchangeability)

$(Z_1, \dots, Z_N)$ is **exchangeable** if $(Z_{\pi(1)}, \dots, Z_{\pi(N)}) \overset{d}{=} (Z_1, \dots, Z_N)$ for every permutation π.

I.i.d. variables are exchangeable; so are equicorrelated normals $N_N(\mu\mathbf{1}, \sigma^2[(1-\rho)I + \rho\mathbf{1}\mathbf{1}'])$. Independent samples from $N(\mu, 1)$ and $N(\mu, 4)$ are *not*.

Theorem 3.3 (Validity of permutation tests)

Let $Z = (Z_1, \dots, Z_N)$ be exchangeable under H_0, let $\mathcal{G}$ be a group of permutations, and let $p(Z) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} I(T(gZ) \geq T(Z))$. Then $P_{H_0}(p(Z) \leq \alpha) \leq \alpha$ for every α, with equality when $\alpha|\mathcal{G}|$ is an integer and there are no ties. The statement holds conditionally on the multiset of observed values, hence also unconditionally.

Proof. Let U be uniform on $\mathcal{G}$, independent of Z; $UZ \overset{d}{=} Z$. Since $\mathcal{G}$ is a group, $\{gUZ : g \in \mathcal{G}\} = \{gZ\}$, so $T(UZ)$ is a uniform member of the fixed multiset $\{T(gZ)\}$ and lies among its top j (ties counted generously) with probability $\leq j/|\mathcal{G}|$. Hence $P(p(Z) \leq \alpha) = P(p(UZ) \leq \alpha) \leq \alpha$; average over the conditioning. ■

Two samples under $H_0 : F = G$: the pooled observations are i.i.d., and the re-assignments are the $\binom{N}{m}$ splits (each from $m!\,n!$ permutations). This is **conditional inference**: values fixed, labels random.

3.4 Random (Monte Carlo) permutation tests

When $\binom{N}{m}$ is too large, draw M random relabelings, compute $T_1, \dots, T_M$, set $T_0 = T_{\text{obs}}$, and use

$$\hat{p} = \frac{1 + \sum_{i=1}^{M} I(T_i \geq T_{\text{obs}})}{M + 1}. \tag{3.2}$$

Proposition 3.4

Under H_0, $P(\hat{p} \leq \alpha) \leq \alpha$ for every M. Moreover $\hat{p}$ estimates the exact permutation p-value with standard error about $\sqrt{p(1-p)/M}$.

Proof. Given the multiset, $T_0, \dots, T_M$ are exchangeable under H_0 (the observed labeling is itself a uniform draw), so the rank of T_0 is uniform (ties broken against rejection) and $\hat{p} \leq j/(M+1)$ iff T_0 is among the j largest. Second claim: $\hat{p} \approx$ a mean of M Bernoulli(p) indicators. ■

The "+1" makes the Monte Carlo test exactly valid.

Random relabeling. Draw a uniform permutation of $1, \dots, N$ and assign group 1 to its first m positions; each m-subset arises from $m!\,n!$ of the $N!$ permutations, so has probability $1/\binom{N}{m}$.

Alternatives and p-values. One-sided alternatives to $H_0 : F = G$ are stochastic orderings, $G(x) \le F(x)$ for all x or the reverse ($\Delta > 0$ or $\Delta < 0$ in a shift model). For $D = \bar{Y} - \bar{X}$ with values $d_0 = D_{\text{obs}}, d_1, \ldots, d_M$:

$$p_{\text{upper}} = \frac{\#\{d_i \ge d_0\}}{M+1}, \qquad p_{\text{lower}} = \frac{\#\{d_i \le d_0\}}{M+1}, \qquad p_{\text{two}} = \frac{\#\{|d_i| \ge |d_0|\}}{M+1}.$$

With $M = 1999$, the standard error at $p = 0.05$ is about 0.005.

3.5 Other permutation tests

One sample: sign flipping. If $X_i - \theta_0$ are independent and symmetric about 0 under H_0, all 2^n sign choices give the same distribution. Flipping signs with $\sum I(X_i > \theta_0)$ gives the sign test, with $\sum V_i R_i$ the signed-rank test, with the t-statistic a symmetric-F t-test.

2×2 tables: Fisher's exact test. For two binary variables and H_0: independence, permute one variable. With margins fixed, the (exposed, diseased) count D is hypergeometric:

$$P(D = d) = \binom{m}{d}\binom{N-m}{k-d} \Big/ \binom{N}{k},$$

where m is the number diseased and k the number exposed among N.

Example 3.2

$N = 20$ subjects, $m = 10$ diseased, $k = 6$ exposed, and 5 of the exposed are diseased. Then $P(D \ge 5) = [\binom{10}{5}\binom{10}{1} + \binom{10}{6}\binom{10}{0}]/\binom{20}{6} = (2520 + 210)/38760 = 0.070$.

Summary. Permutation tests need only exchangeability under H_0, allow any statistic and give exact p-values; but they test only hypotheses making the data exchangeable (not "equal means, unequal variances"), and intervals require inverting many tests.

3.6 Selected exercises

Motivation. Monte Carlo error of $\hat{p}$, 2×2 statistics as Fisher tests, and moments of the permutation distribution.

Exercise 3.6 (Monte Carlo error of $\hat{p}$)

A random permutation test with $M = 999$ relabelings gave $\hat{p} = 0.01$. Give an approximate 99% confidence interval for the exact permutation p-value p. How much shorter would it be with $M = 1999$?

Solution.

Step 1. Given the data, each relabeling is uniform over the K splits, so $P(T_i \ge T_{\text{obs}}) = p$ by (3.1); independence gives $\sum_{i=1}^{M} I(T_i \ge T_{\text{obs}}) \sim B(M, p)$, and $\hat{p}$ in (3.2) differs from that proportion by $O(1/M)$. So $\hat{p} \dot{\sim} N(p, p(1-p)/M)$.

Step 2. With $z_{0.005} = 2.576$ and $\hat{p}$ in the standard error,

$$\hat{p} \pm 2.576\sqrt{\frac{\hat{p}(1-\hat{p})}{M}} = 0.01 \pm 2.576\sqrt{\frac{0.0099}{999}} = 0.01 \pm 0.0081 = (0.0019,\ 0.0181).$$

Step 3. With $M = 1999$ the half-width is $2.576\sqrt{0.0099/1999} = 0.0057$: (0.0043, 0.0157). Width scales as $M^{-1/2}$, so doubling M shortens it by $1 - \sqrt{999/1999} \approx 29\%$; halving it needs $4M$. ■

Exercise 3.7–3.8 (χ^2 and odds ratio as permutation statistics)

In the 2×2 setting of Section 3.5 (margins fixed: N subjects, m diseased, k exposed, D = number exposed and diseased), show:
(a) Pearson's χ^2 statistic is an increasing function of $|D - km/N|$, so the permutation test based on χ^2 is the two-sided Fisher test that orders tables by $|D - ED|$;
(b) the sample odds ratio is an increasing function of D, so the permutation test based on it is the one-sided Fisher test.
Apply both to Example 3.2.

Solution. Cells: $a = D$, $b = k - D$, $c = m - D$, $d = N - m - k + D$; row totals $k, N - k$, column totals $m, N - m$.

(a)

Step 1. Expected counts are $E_a = km/N$, $E_b = k(N - m)/N$, $E_c = (N - k)m/N$, $E_d = (N - k)(N - m)/N$, so

$$a - E_a = d - E_d = D - \frac{km}{N}, \qquad b - E_b = c - E_c = -\Big(D - \frac{km}{N}\Big).$$

Step 2. Hence $\chi^2 = \sum(O - E)^2/E = (D - km/N)^2 \sum 1/E$, and

$$\sum \frac{1}{E} = \frac{N}{k\,m\,(N-k)(N-m)}\big[(N-k)(N-m)+(N-k)m+k(N-m)+km\big] = \frac{N^3}{k\,m\,(N-k)(N-m)}.$$

The constant is the same for every permutation (fixed margins), so χ^2 is increasing in $|D - km/N|$, and with $ED = km/N$ the p-value is $P(|D - km/N| \geq |d_{\text{obs}} - km/N|)$.

(b) Where $b, c > 0$, $\text{OR} = \frac{ad}{bc} = \frac{D}{m-D} \cdot \frac{N-m-k+D}{k-D}$, each factor nonnegative increasing over positive decreasing, so OR increases in D (strictly where positive). At the ends OR is 0 (a or $d = 0$) or $+\infty$ (b or $c = 0$). So "$\text{OR} \geq \text{OR}_{\text{obs}}$" is $D \geq d_{\text{obs}}$: the one-sided Fisher test.

In Example 3.2, $km/N = 3$, $d = 5$, and $m = N - m = 10$ makes $P(D = d)$ symmetric about 3. Odds ratio: $p = P(D \geq 5) = 0.070$; χ^2: $p = P(D \geq 5) + P(D \leq 1) = 0.141$. ■

Exercise 3.9 (Moments of the permutation distribution of $\bar{Y} - \bar{X}$)

Let $z_1, \ldots, z_N$ be the pooled observations, $S^2 = \frac{1}{N-1}\sum_k (z_k - \bar{z})^2$, and let $D = \bar{Y} - \bar{X}$ be computed for a uniformly random split into groups of sizes m (X) and n (Y).
(a) Show that $ED = 0$ and $\operatorname{Var} D = NS^2/(mn)$.
(b) Let $Z = D/\sqrt{NS^2/(mn)}$ and let T be the pooled two-sample t-statistic. Show that $Z^2 = (N - 1)T^2/(N - 2 + T^2)$. Hence Z is an increasing function of T, and $Z/T \to 1$ as $N \to \infty$ with T fixed.

Solution [Supplement].

(a) $\bar{Y}$ is the mean of an SRS of size n from $\{z_k\}$, population variance $\sigma^2 = (N - 1)S^2/N$. By Theorem 4.2,

$$E\bar{Y} = \bar{z}, \qquad \operatorname{Var}\bar{Y} = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{mS^2}{nN}.$$

Since $m\bar{X} + n\bar{Y} = N\bar{z}$, $D = \frac{N}{m}(\bar{Y} - \bar{z})$, so $ED = 0$ and $\operatorname{Var} D = \frac{N^2}{m^2} \cdot \frac{mS^2}{nN} = \frac{NS^2}{mn}$.

(b) From the proof of Proposition 3.2, $(N-1)S^2 = SST = (N-2)s_p^2 + A$ with $A = \frac{mn}{N}D^2$, and $T^2 = A/s_p^2$. Therefore

$$Z^2 = \frac{A}{S^2} = \frac{(N-1)A}{(N-2)s_p^2 + A} = \frac{(N-1)T^2}{N-2+T^2}.$$

Both Z and T have the sign of D; thus $Z = g(T)$ with $g(t) = t\sqrt{(N-1)/(N-2+t^2)}$, where $g'(t) = \sqrt{N-1}\,(N-2)(N-2+t^2)^{-3/2} > 0$ and $g(t)/t \to 1$ as $N \to \infty$.

Hence permutation and normal-theory p-values nearly coincide in large samples. ■